

# Tshingombe Tshitadi Fiston

Professional Researcher, Curriculum Author & Electrical Engineering Facilitator

Email: tshingombefiston@gmail.com | Affiliation: Sci-Bono Discovery Centre | Specialization: Electrical Engineering & TVET Systems

## PROFESSIONAL EXECUTIVE SUMMARY

Highly accomplished and multidisciplinary Researcher, Curriculum Author, and Engineering Facilitator with over 14 years of longitudinal data and operational experience in Technical and Vocational Education and Training (TVET). Recognized expert in the design of auditable, career-aligned technical education frameworks, N1-N6 Electrotechnics, Diesel Trade theory, and advanced microgrid modeling. Deeply specialized in South African and international vocational regulatory frameworks, including SAQA, NQF, DHET, QCTO compliance, and cross-border educational alignment between South Africa and the Democratic Republic of the Congo (DRC).

## CORE COMPETENCIES & TECHNICAL FRAMEWORKS

Electrical Engineering Master Curriculum

N1-N6 Electrotechnics

Diesel Trade Frameworks

SAQA / NQF / DHET / QCTO Compliance

Microgrid & Hybrid Energy Systems

VBA & Technical Automation

Procurement Science & Tender Evaluation

ISO/IEC 27001 Implementation

## PROFESSIONAL EXPERIENCE & AFFILIATIONS

### Facilitator, Electrical Engineering & Curriculum Development

*Sci-Bono Discovery Centre | Ongoing*

- Spearheading technical facilitation in electrical engineering, N1-N6 Electrotechnics, and industrial trade frameworks.
- Authoring comprehensive curriculum documents and auditable technical education modules aligned with rigorous regulatory standards.
- Managing technical bid submissions, procurement compliance, and site verifications for advanced engineering educational equipment.

### Researcher & Curriculum Author

*Longitudinal Educational Systems & Framework Design | 14+ Years*

- Engineered credentialed engineering frameworks and technical documentation automation systems using VBA logic, logigrams, and algorigrams.
- Developed specialized qualification translation frameworks harmonizing South African (SAQA) and DRC educational structures.
- Conducted advanced research in microgrid modeling, smart grid AI integration, and crime information management systems within public safety sectors.

## CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **Assessment Development Fundamentals & SQA Qualification Delivery:** Completed advanced coursework in vocational assessment, marking, and Scottish Qualifications Authority (SQA) frameworks.
- **Information Security & Compliance:** Active engagement in ISACA governance frameworks, chapter compliance, and ISO/IEC 27001 implementation methodologies.
- **Advanced Energy Systems:** Specialized synthesis in Microgrid Modeling, Analysis, and Smart Grid AI Integration.

## **MOTIVATION LETTER / STATEMENT OF PURPOSE**

As a dedicated educator, curriculum author, and electrical engineering specialist, my career has been defined by a relentless pursuit of technical excellence, regulatory rigor, and institutional innovation. Over the past 14 years, I have bridged the gap between theoretical electrical engineering and practical vocational execution, designing robust curricula that empower the next generation of technicians and engineers.

My work at the intersection of TVET frameworks (SAQA, DHET, QCTO) and advanced energy systems—such as hybrid microgrids and smart grid automation—reflects my commitment to future-proofing technical education across the African continent. Whether authoring N1-N6 Electrotechnics modules, establishing cross-border qualification equivalencies, or streamlining procurement and audit trails through automated software logic, I bring unyielding dedication, strategic insight, and meticulous precision to every professional undertaking.

I welcome opportunities to collaborate on visionary educational initiatives, engineering framework designs, and technological advancements that drive sustainable industrial growth.